GEOMETRIC CONNECTEDNESS TESTED OVER A SEPARABLE CLOSURE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 2376-2384, immediately following the statement at 2368-2374.

Editorial note. The reverse (only-if) direction is the definition of geometric connectedness read on overline{k}/k; the author text starts from the forward assumption.

Editorial note. One same-chapter lemma stub is recorded in assembly.external_references.

Allowed standard prerequisites (editorial).

- Connectedness over separably closed extensions. Lemma lemma-separably-closed-field-connected-components of this chapter, cited at proof lines 2379-2380. X_{overline{k}'} stays connected because overline{k}' is separably closed over k.
- Surjective morphisms reflect connectedness of the source onto fibres. standard fact invoked at proof lines 2381-2383 (X_{overline{k}'} -> X_{k'} surjective). A connected space surjecting onto X_{k'} forces X_{k'} connected.

Source reference. Original lemma 0A12: sources/stacks-spaces-over-fields/pinned-original-source.tex, source lines 2350–2350. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let k be a field. Let X be an algebraic space over k. Let $\overline{k}$ be a separable algebraic closure of k. Then X is geometrically connected if and only if the base change $X_{\overline{k}}$ is connected.*

Proof. Assume $X_{\overline{k}}$ is connected. Let k'/k be a field extension. There exists a field extension $\overline{k}'/\overline{k}$ such that k' embeds into $\overline{k}'$ as an extension of k. By Lemma 0A12 we see that $X_{\overline{k}'}$ is connected. Since $X_{\overline{k}'} \to X_{k'}$ is surjective we conclude that $X_{k'}$ is connected as desired. □

Date: stacks-project:spaces-over-fields.tex:local-snapshot-20261012; GNU FDL 1.2 or later.